Binomial Distribution

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Conditions for a binomial distribution**

State the four conditions required for a random variable to follow a binomial distribution.

(1) A fixed number of trials n. (2) Each trial has only two outcomes (success or failure). (3) The probability of success p is constant for every trial. (4) The trials are independent of each other.

2 **[Foundational]** • **Target: Conditions for a binomial distribution**

A survey asks 50 people whether they support a new policy, and X is the number who say yes, assuming each person's response is independent with the same probability of support. Explain why X follows a binomial distribution.

X follows a binomial distribution because there is a fixed number of trials ($n = 50$), each trial (person) has two outcomes (yes or no), the probability of a 'yes' response is constant across all people, and the responses are independent of each other — satisfying all four binomial conditions.

3 **[Foundational]** • **Target: Conditions for a binomial distribution**

A researcher rolls a die 15 times and counts the number of times a 3 appears. Explain why this count follows a binomial distribution.

This count follows a binomial distribution because there is a fixed number of trials ($n = 15$), each roll has two outcomes (rolling a 3 or not), the probability of rolling a 3 is constant ($\frac{1}{6}$) on every roll, and the rolls are independent of each other — satisfying all four binomial conditions.

4 **[Foundational]** • **Target: Calculating binomial probabilities**

A fair coin is tossed 8 times. Find the probability of getting exactly 5 heads, using $P(X = r) = \binom{n}{r} p^r (1-p)^{n-r}$.

$X \sim B(8, 0.5)$. $P(X = 5) = \binom{8}{5}(0.5)^5(0.5)^3 = 56(0.03125)(0.125) = 56(0.00390625) \approx 0.2188$.

5 **[Foundational]** • **Target: Calculating binomial probabilities**

A factory produces items with a defect rate of 10%. In a sample of 6 items, find the probability that exactly 2 are defective.

$X \sim B(6, 0.1)$. $P(X = 2) = \binom{6}{2}(0.1)^2(0.9)^4 = 15(0.01)(0.6561) \approx 0.0984$.

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6 [Foundational] • Target: Calculating binomial probabilities

A fair coin is tossed 6 times. Find the probability of getting exactly 4 heads.

$X \sim B(6, 0.5)$. $P(X = 4) = \binom{6}{4}(0.5)^4(0.5)^2 = 15(0.0625)(0.25) = 0.234375$.

7 [Foundational] • Target: Calculating binomial probabilities

A factory produces items with a defect rate of 20%. In a sample of 5 items, find the probability that exactly 1 is defective.

$X \sim B(5, 0.2)$. $P(X = 1) = \binom{5}{1}(0.2)^1(0.8)^4 = 5(0.2)(0.4096) = 0.4096$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Cumulative binomial probabilities

A biased coin has $P(\text{heads}) = 0.3$. It is tossed 5 times. Find $P(X \leq 1)$, where X is the number of heads.

$P(X = 0) = \binom{5}{0}(0.3)^0(0.7)^5 = 0.16807$. $P(X = 1) = \binom{5}{1}(0.3)^1(0.7)^4 = 5(0.3)(0.2401) = 0.36015$.
$P(X \leq 1) = 0.16807 + 0.36015 \approx 0.5282$.

9 [Intermediate] • Target: Cumulative binomial probabilities

Using the same distribution $X \sim B(5, 0.3)$, find $P(X \geq 2)$.

$P(X \geq 2) = 1 - P(X \leq 1) = 1 - 0.5282 \approx 0.4718$.

10 [Intermediate] • Target: Cumulative binomial probabilities

A biased coin has $P(\text{heads}) = 0.2$. It is tossed 4 times. Find $P(X \leq 1)$, where X is the number of heads.

$P(X = 0) = \binom{4}{0}(0.2)^0(0.8)^4 = 0.4096$. $P(X = 1) = \binom{4}{1}(0.2)^1(0.8)^3 = 4(0.2)(0.512) = 0.4096$.
$P(X \leq 1) = 0.4096 + 0.4096 = 0.8192$.

11 [Intermediate] • Target: Mean and variance of a binomial distribution

A random variable $X \sim B(40, 0.25)$. Find the mean and variance, using $E(X) = np$ and $\text{Var}(X) = np(1 - p)$.

$E(X) = np = 40(0.25) = 10$. $\text{Var}(X) = np(1 - p) = 40(0.25)(0.75) = 7.5$.

12 [Intermediate] • Target: Mean and variance of a binomial distribution

A random variable $X \sim B(60, 0.2)$. Find the mean and variance.

$E(X) = np = 60(0.2) = 12$. $\text{Var}(X) = np(1 - p) = 60(0.2)(0.8) = 9.6$.

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13 [Intermediate] • Target: Mean and variance of a binomial distribution

A random variable $X \sim B(80, 0.1)$. Find the mean and variance.

$E(X) = np = 80(0.1) = 8$. $\text{Var}(X) = np(1-p) = 80(0.1)(0.9) = 7.2$.

14 [Intermediate] • Target: Mean and variance of a binomial distribution

A random variable $X \sim B(50, 0.6)$. Find the mean and variance.

$E(X) = np = 50(0.6) = 30$. $\text{Var}(X) = np(1-p) = 50(0.6)(0.4) = 12$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Finding an unknown parameter • Cross-Lesson: 14.01 Energy, Work & Power

A random variable $X \sim B(n, 0.4)$ has mean $E(X) = 20$. Find n.

$E(X) = np$: $20 = n(0.4)$, so $n = 50$.

16 [Challenging] • Target: Finding an unknown parameter • Cross-Lesson: 15.01 Representation of Data

A random variable $X \sim B(25, p)$ has mean $E(X) = 10$. Find p.

$E(X) = np$: $10 = 25p$, so $p = 0.4$.

17 [Challenging] • Target: Finding an unknown parameter • Cross-Lesson: 16.01 Permutations & Combinations

A random variable $X \sim B(n, 0.25)$ has variance $\text{Var}(X) = 9$. Find n.

$\text{Var}(X) = np(1-p)$: $9 = n(0.25)(0.75) = 0.1875n$, so $n = \frac{9}{0.1875} = 48$.

18 [Challenging] • Target: Finding an unknown parameter • Cross-Lesson: 17.01 Discrete Random Variables

A random variable $X \sim B(n, 0.5)$ has variance $\text{Var}(X) = 12.5$. Find n.

$\text{Var}(X) = np(1-p)$: $12.5 = n(0.5)(0.5) = 0.25n$, so $n = \frac{12.5}{0.25} = 50$.

19 [Challenging] • Target: Using the binomial distribution to model a real scenario • Cross-Lesson: 19.01 Normal Distribution

A multiple-choice quiz has 10 questions, each with 4 options, only one of which is correct. A student guesses randomly on every question. Find the probability the student gets at least 8 questions correct.

$X \sim B(10, 0.25)$. $P(X = 8) = \binom{10}{8}(0.25)^8(0.75)^2 \approx 45(0.0000153)(0.5625) \approx 0.000387$.

$P(X = 9) = \binom{10}{9}(0.25)^9(0.75)^1 \approx 10(0.00000381)(0.75) \approx 0.0000286$. $P(X = 10) = (0.25)^{10} \approx 0.00000095$.

$P(X \geq 8) \approx 0.000387 + 0.0000286 + 0.00000095 \approx 0.000417$.

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20 [Challenging] • Target: Using the binomial distribution to model a real scenario • Cross-Lesson: 2.01 Simultaneous Equations & Inequalities

A basketball player has a 70% free-throw success rate. Find the probability that the player makes exactly 3 out of 5 free throws.

$X \sim B(5, 0.7)$. $P(X = 3) = \binom{5}{3}(0.7)^3(0.3)^2 = 10(0.343)(0.09) = 0.3087$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Binomial Criteria & Single Probabilities

Which of the following is NOT a required condition for a variable X to have a binomial distribution $B(n, p)$?

a. The probability of success changes with each trial ✓

b. There is a fixed number of trials n

c. Each trial results in one of only two outcomes (success or failure)

d. Trials are independent of one another

In a binomial distribution, the probability of success p must remain strictly constant across all trials.

22 [Foundational] • Target: Binomial Criteria & Single Probabilities

If $X \sim B(6, 0.3)$, calculate $P(X = 3)$ correct to 4 decimal places.

a. 0.1852 ✓ **b.** 0.2541

c. 0.3087 **d.** 0.1176

$P(X = 3) = \binom{6}{3}(0.3)^3(0.7)^3 = 20 \times 0.027 \times 0.343 = 0.18522 \approx 0.1852$.

23 [Foundational] • Target: Binomial Criteria & Single Probabilities

If $X \sim B(5, 0.4)$, find $P(X = 0)$.

a. 0.0778 ✓ **b.** 0.0102

c. 0.0625 **d.** 0.1000

$P(X = 0) = (1 - 0.4)^5 = (0.6)^5 = 0.07776 \approx 0.0778$.

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24 [Foundational] • Target: Cumulative Binomial Probabilities

If $X \sim B(4, 0.5)$, find $P(X \geq 2)$.

a. $\frac{11}{16} = 0.6875$ ✓

b. $\frac{1}{2} = 0.5000$

c. $\frac{5}{16} = 0.3125$

d. $\frac{3}{8} = 0.3750$

$P(X < 2) = P(0) + P(1) = \frac{1}{16} + \frac{4}{16} = \frac{5}{16}$. Then $P(X \geq 2) = 1 - 5/16 = 11/16 = 0.6875$.

25 [Foundational] • Target: Cumulative Binomial Probabilities

A manufacturer knows that 5% of lightbulbs are defective. In a random sample of 20 bulbs, what is the probability that at most 1 is defective?

a. 0.7358 ✓

b. 0.6824

c. 0.8245

d. 0.9128

$X \sim B(20, 0.05)$. $P(X \leq 1) = P(0) + P(1) = (0.95)^{20} + 20(0.05)(0.95)^{19} = 0.358486 + 0.377354 = 0.73584 \approx 0.7358$.

26 [Foundational] • Target: Cumulative Binomial Probabilities

A fair coin is tossed 6 times. Find the probability of obtaining at least 5 heads.

a. $\frac{7}{64} \approx 0.1094$ ✓

b. $\frac{6}{64} \approx 0.0938$

c. $\frac{1}{64} \approx 0.0156$

d. $\frac{11}{64} \approx 0.1719$

$P(X \geq 5) = P(5) + P(6) = \binom{6}{5}(1/2)^6 + \binom{6}{6}(1/2)^6 = \frac{6+1}{64} = \frac{7}{64} \approx 0.1094$.

27 [Foundational] • Target: Mean & Variance of the Binomial Distribution

A random variable $X \sim B(48, 0.25)$. Find its mean and standard deviation.

a. **Mean** $= 12, \sigma = 3$ ✓

b. Mean $= 12, \sigma = 9$

c. Mean $= 16, \sigma = 4$

d. Mean $= 12, \sigma = \sqrt{3}$

Mean $\mu = np = 48(0.25) = 12$. Variance $\sigma^2 = np(1-p) = 48(0.25)(0.75) = 9$. Standard deviation $\sigma = \sqrt{9} = 3$.

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28 [Foundational] • Target: Mean & Variance of the Binomial Distribution

For a binomial distribution $X \sim B(n, p)$, the mean is 6 and the variance is 3.6. Find n and p.

a. $n = 15, p = 0.4$ ✓

b. $n = 20, p = 0.3$

c. $n = 12, p = 0.5$

d. $n = 24, p = 0.25$

$\sigma^2 = \mu(1-p) \Rightarrow 3.6 = 6(1-p) \Rightarrow 1 - p = 0.6 \Rightarrow p = 0.4$. Then $n = 6/0.4 = 15$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Mean & Variance of the Binomial Distribution

What is the maximum possible variance of a binomial distribution with $n = 40$ trials?

a. **10** ✓

b. 20

c. 5

d. 40

Variance $= np(1-p)$. The quadratic $p(1-p)$ is maximized at $p = 0.5$, giving maximum variance $= 40(0.5)(0.5) = 10$.

30 [Intermediate] • Target: Contextual Modeling & Problem Solving

A treatment is successful for 80% of patients. If 10 patients are treated, what is the most likely number of successful outcomes (the mode)?

a. **8** ✓

b. 7

c. 9

d. 10

The mean is $np = 10(0.8) = 8$. For integer $(n+1)p = 11(0.8) = 8.8$, the mode is the integer part $\lfloor 8.8 \rfloor = 8$.

31 [Intermediate] • Target: Contextual Modeling & Problem Solving

How many times must a fair die be rolled so that the probability of getting at least one 6 is greater than 0.9 ?

a. **At least 13 rolls** ✓

b. At least 10 rolls

c. At least 6 rolls

d. At least 18 rolls

$P(X \geq 1) = 1 - (5/6)^n > 0.9 \Rightarrow (5/6)^n < 0.1 \Rightarrow n \ln(5/6) < \ln(0.1) \Rightarrow n > \frac{\ln(0.1)}{\ln(5/6)} = \frac{-2.3026}{-0.1823} \approx 12.63$. Least integer is 13.

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32 [Intermediate] • Target: Contextual Modeling & Problem Solving

In an exam, 70% of candidates pass. Out of 8 randomly chosen candidates, what is the probability that exactly 6 pass, given that at least 5 pass?

a. ≈ 0.380 ✓

b. ≈ 0.520

c. ≈ 0.296

d. ≈ 0.420

$P(6 \mid \geq 5) = \frac{P(X=6)}{P(X \geq 5)}$. Calculating with $B(8, 0.7)$: $P(5) = 0.2541, P(6) = 0.2965, P(7) = 0.1977, P(8) = 0.0576$. Sum $= 0.8059$. Ratio $= 0.2965/0.8059 \approx 0.368$ (approx 0.38).

33 [Intermediate] • Target: Conditions for a binomial distribution

A factory produces light bulbs. The probability that a randomly selected light bulb is defective is p. The number of defective light bulbs in a random sample of n bulbs follows a binomial distribution $X \sim B(n, p)$. It is known that the mean number of defective bulbs in a sample of $n = 25$ is 2. \\ (i) Calculate the value of p. \\ (ii) For a sample of $n = 50$ bulbs, calculate the probability of finding exactly k defective bulbs, where k is the smallest integer such that $P(X \leq k) > 0.9$. \\ (iii) If a sample of $n = 50$ bulbs is taken, find the smallest value of n' such that the probability of observing at least one defective bulb is greater than 0.999 .

a. 0.04

b. **0.08** ✓

c. 0.12

d. 0.16

Part (i): For a binomial distribution $X \sim B(n, p)$, the mean is given by $E(X) = np$. \\ We are given $n = 25$ and $E(X) = 2$. \\ So, $25p = 2$, which gives $p = \frac{2}{25} = 0.08$. \\ Part (ii): Now, $n = 50$ and $p = 0.08$. The number of defective bulbs follows $X \sim B(50, 0.08)$. \\ We need to find the smallest integer k such that $P(X \leq k) > 0.9$. \\ We can use a binomial cumulative distribution table or a calculator. \\ Let's calculate a few values: $P(X \leq 3) = P(X = 0) + P(X = 1) + P(X = 2) + P(X = 3)$ Using.

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34 **[Intermediate]** • **Target: Calculating binomial probabilities**

The scores on a standardized test are normally distributed with a mean of 100 and a standard deviation of 15. A new scoring algorithm is implemented, and the scores are transformed using the rule $Y = aX + b$, where X is the original score and Y is the new score. The new distribution of scores, Y, also follows a normal distribution with a mean of 120 and a standard deviation of 22.5. Calculate the value of a and the value of b.

a. **a = 1.5, b = -30** ✓

b. a = -1.5, b = 270

c. a = 1.5, b = 30

d. a = 0.67, b = 53

Let $X \sim N(\mu_X, \sigma_X^2)$ and $Y \sim N(\mu_Y, \sigma_Y^2)$. We are given $\mu_X = 100$, $\sigma_X = 15$, $\mu_Y = 120$, and $\sigma_Y = 22.5$. The linear transformation is given by $Y = aX + b$. We know that for a linear transformation $Y = aX + b$, the mean of Y is $\mu_Y = a\mu_X + b$ and the standard deviation of Y is $\sigma_Y = |a|\sigma_X$. From the standard deviation relationship: $\sigma_Y = |a|\sigma_X$ $22.5 = |a| \times 15$ $|a| = \frac{22.5}{15} = 1.5$ Now consider the mean relationship.

35 **[Intermediate]** • **Target: Cumulative binomial probabilities**

A fair six-sided die is rolled 4 times. Let X be the number of times a 6 is obtained. The possible values for X are 0, 1, 2, 3, and 4. The probability of obtaining a 6 on any given roll is $p = \frac{1}{6}$. The number of trials is $n = 4$. This scenario can be modelled by a binomial distribution, denoted as $X \sim B(n, p)$. Which of the following represents the correct parameters for this binomial distribution?

a. $B(4, 1/6)$ ✓

b. $B(6, 1/4)$

c. $B(1, 4/6)$

d. $B(4, 4/6)$

The problem describes a situation with a fixed number of independent trials (rolling a die 4 times), where each trial has only two possible outcomes (obtaining a 6 or not obtaining a 6), and the probability of success (obtaining a 6) is constant for each trial. In a binomial distribution, denoted as $X \sim B(n, p)$, n represents the number of trials, and p represents the probability of success on a single trial. From the question: - The number of times the die is rolled is 4, so $n = 4$. - The probability of o.

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36 **[Intermediate]** • **Target: Mean and variance of a binomial distribution**

Cards are dealt from a single shuffled deck without replacement; X is the number of aces dealt. Is X binomially distributed?

a. No — trials are not independent / probability is not constant ✓

b. Yes — but only if $n > 30$

c. Yes — fixed, independent trials with constant success probability

d. No — the number of trials is not fixed

The conditions for a binomial distribution (fixed independent trials with constant probability) are not satisfied here.

37 **[Intermediate]** • **Target: Finding an unknown parameter**

A fair coin is tossed 3 times. Let X be the number of heads obtained. What is the probability that X = 2?

a. $\frac{1}{8}$

b. $\frac{3}{8}$ ✓

c. $\frac{1}{2}$

d. $\frac{7}{8}$

This is a binomial distribution scenario. The number of trials is $n = 3$ (the number of coin tosses). The probability of success (getting a head) in a single trial is $p = \frac{1}{2}$, and the probability of failure (getting a tail) is $q = 1 - p = \frac{1}{2}$. We want to find the probability of getting exactly $k = 2$ heads. The binomial probability formula is $P(X = k) = \binom{n}{k} p^k q^{n-k}$. Substituting the values: $P(X = 2) = \binom{3}{2} (\frac{1}{2})^2 (\frac{1}{2})^{3-2}$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Using the binomial distribution to model a real scenario • Cross-Lesson: 17.01 Discrete Random Variables

The weights of bags of sugar are normally distributed with a mean of 1000 g and a standard deviation of 5 g. Let X be the weight of a randomly selected bag of sugar. Calculate the probability that a bag of sugar weighs between 990 g and 1010 g.

a. **0.9545 ✓**

b. 0.6827

c. 0.9772

d. 0.0228

Let X be the weight of a randomly selected bag of sugar. We are given that $X \sim N(\mu = 1000, \sigma^2 = 5^2)$. We want to find $P(990 < X < 1010)$. First, we convert the weights to z-scores: For $X = 990$, $z = \frac{990 - 1000}{5} = \frac{-10}{5} = -2$. For $X = 1010$, $z = \frac{1010 - 1000}{5} = \frac{10}{5} = 2$. So we need to find $P(-2 < Z < 2)$, where Z is the standard normal distribution. This probability can be found using a standard normal distribution table or calculator. The area to the left of $z = 2$ is.

39 [Challenging] • Target: Distinguishing binomial situations from non-binomial situations • Cross-Lesson: 19.01 Normal Distribution

A factory tests 30 independent components, each with a fixed probability of being defective; X is the number of defective components. Is X binomially distributed?

a. **Yes — fixed, independent trials with constant success probability ✓**

b. No — the number of trials is not fixed

c. No — trials are not independent / probability is not constant

d. Yes — but only if $n > 30$

The trials are fixed in number, independent, and each has the same probability of success, so the conditions for a binomial distribution are satisfied.

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40 [Challenging] • Target: A well-designed word problem: quality control sampling • Cross-Lesson: 2.01 Simultaneous Equations & Inequalities

A die is rolled 15 times; X is the number of sixes. Is X binomially distributed?

a. No — trials are not independent / probability is not constant

b. Yes — but only if $n > 30$

c. **Yes — fixed, independent trials with constant success probability ✓**

d. No — the number of trials is not fixed

The trials are fixed in number, independent, and each has the same probability of success, so the conditions for a binomial distribution are satisfied.

41 [Challenging] • Target: Binomial Criteria & Single Probabilities • Cross-Lesson: 3.01 Graphs & Transformations

A fair coin is tossed 20 times; X is the number of heads. Is X binomially distributed?

a. No — trials are not independent / probability is not constant

b. Yes — but only if $n > 30$

c. **Yes — fixed, independent trials with constant success probability ✓**

d. No — the number of trials is not fixed

The trials are fixed in number, independent, and each has the same probability of success, so the conditions for a binomial distribution are satisfied.

42 [Challenging] • Target: Cumulative Binomial Probabilities • Cross-Lesson: 4.01 Coordinate Geometry

A survey asks 50 randomly and independently selected people the same yes/no question; X is the number who say yes. Is X binomially distributed?

a. No — trials are not independent / probability is not constant

b. No — the number of trials is not fixed

c. **Yes — fixed, independent trials with constant success probability ✓**

d. Yes — but only if $n > 30$

The trials are fixed in number, independent, and each has the same probability of success, so the conditions for a binomial distribution are satisfied.

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43 [Challenging] • Target: Mean & Variance of the Binomial Distribution • Cross-Lesson: 5.01 Trigonometry

Given $X \sim B(73, 17/20)$, find $E(X)$.

a. 1221/20

b. 1251/20

c. 1261/20

d. **1241/20** ✓

Applying Binomial Distribution principles, $E(X) = np = 73 \times 17/20 = 1241/20$ Therefore, the verified correct option is 1241/20.

44 [Challenging] • Target: Contextual Modeling & Problem Solving • Cross-Lesson: 6.01 Differentiation & Integration

Given $X \sim B(39, 11/20)$, find $E(X)$.

a. 409/20

b. **429/20** ✓

c. 439/20

d. 449/20

Applying Binomial Distribution principles, $E(X) = np = 39 \times 11/20 = 429/20$ Therefore, the verified correct option is 429/20.

45 [Challenging] • Target: Conditions for a binomial distribution • Cross-Lesson: 7.01 Statistics & Probability

Given $X \sim B(38, 13/20)$, find $E(X)$.

a. **247/10** ✓

b. 257/10

c. 237/10

d. 126/5

Applying Binomial Distribution principles, $E(X) = np = 38 \times 13/20 = 247/10$ Therefore, the verified correct option is 247/10.

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